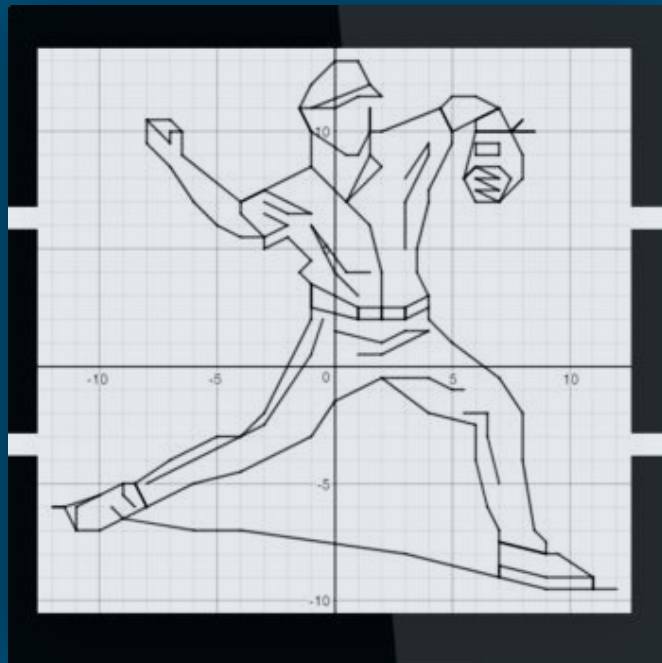


Pitch Prediction Model

Luke Kurta, Francesco Noble, Nathanael Choi



Introduction: Predicting Fastball vs. Offspeed Pitches



The Why: Accurately predicting whether, given the current situation, a pitch will be a fastball or an offspeed pitch gives hitters a major advantage



Pitch selection is based on game context: *e.g. count (balls-strikes), baserunners, score, inning, outs, and pitcher/batter tendencies*



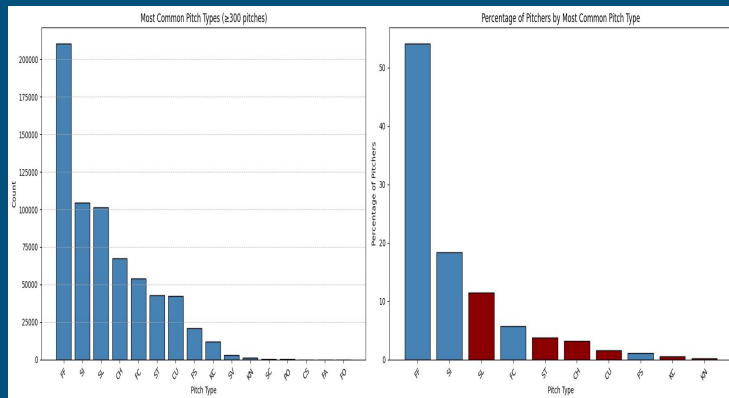
MLB Statcast data: filtered data to have pitchers with 300+ pitches



League-wide trends vs. individual pitcher patterns



We framed the problem as predicting either a fastball or an offspeed pitch using labeled pitch-level data

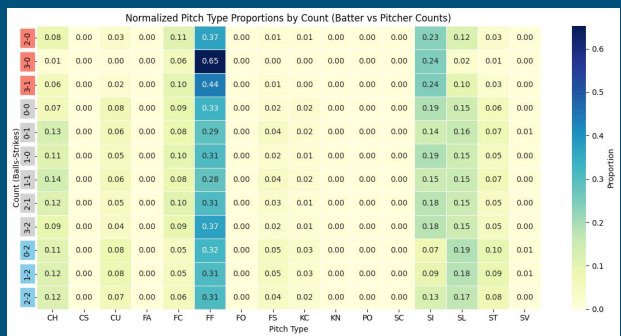


Fastballs dominate pitch selection, with over 200,000 thrown—more than double the count the next most common pitch types. Over 50% of pitchers in 2024 had fastball as their most common pitch

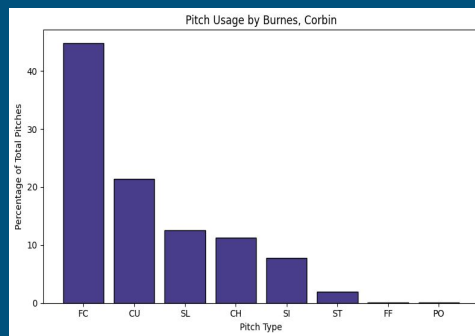
player_name	p_throws	stand	inning	outs_when_up	on_1b	on_2b	on_3b	bat_score	fld_score	pitch_type	balls	strikes
Díaz, Edwin	R	R	9	2		642201		7	8	FF	2	2
Díaz, Edwin	R	R	9	2		642201		7	8	SL	1	2
Díaz, Edwin	R	R	9	2		642201		7	8	SL	1	1
Díaz, Edwin	R	R	9	2		642201		7	8	SL	0	1
Díaz, Edwin	R	R	9	2		642201		7	8	SL	0	0
Díaz, Edwin	R	R	9	1		642201		7	8	SL	1	2
Díaz, Edwin	R	R	9	1		642201		7	8	SL	0	2
Díaz, Edwin	R	R	9	1		642201		7	8	SL	0	2
Díaz, Edwin	R	R	9	1		642201		7	8	FF	0	2
Díaz, Edwin	R	R	9	1	642201			7	8	SL	0	1
Díaz, Edwin	R	R	9	1	642201			7	8	FF	0	0
Díaz, Edwin	R	R	9	1				7	8	SL	3	2
Díaz, Edwin	R	R	9	1				7	8	SL	2	2

Example of raw pitch-level data (each row = one pitch)

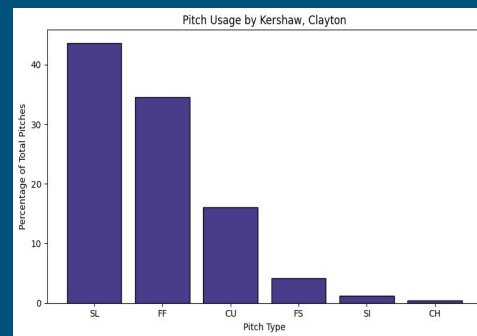
Exploratory Data Analysis: How Pitch Context Shapes Pitch Type



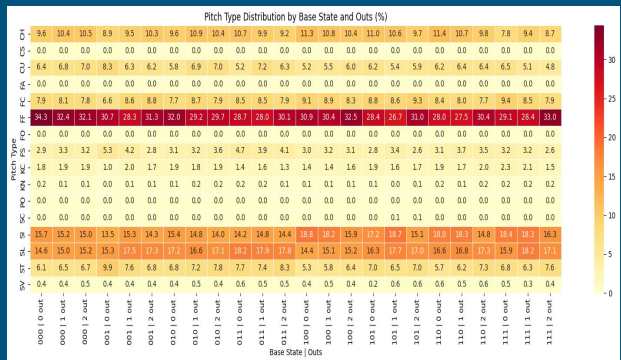
Pitchers lean on fastballs in hitter's count (e.g. 3-0) but throw more offspeed in pitcher's counts (e.g. 0-2)



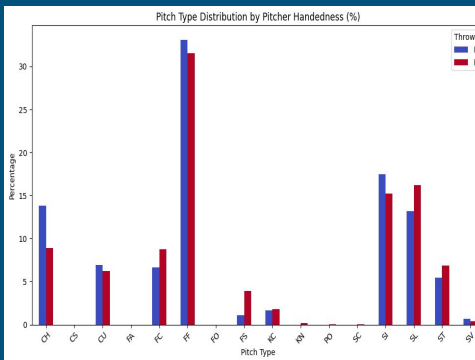
Burnes heavily favors his cutter, a fastball variant, reinforcing that fastballs dominate across the league



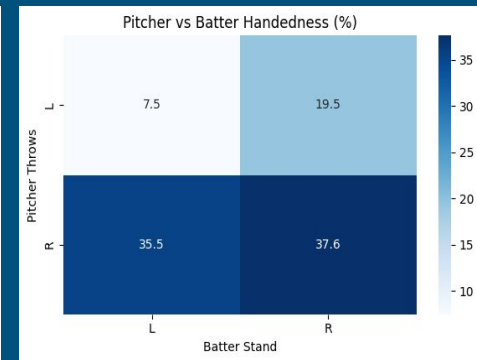
Kershaw prefers slider—an offspeed pitch—across most situations, highlighting unique pitcher tendencies



Offspeed pitches are more common with runners on 2nd or 3rd. With a runner on 1st and one out, fastballs—especially sinkers—are favored to induce ground balls, causing a double play



Pitcher handedness skews the distribution of batter matchups since most MLB pitchers are right-handed, shaping pitch selection patterns. Left-handed pitchers throw more offspeed pitches like changeups and sliders due to the natural movement advantage these pitches create against opposite-handed batters.



Challenges



Class Imbalance

Fastballs are thrown far more than offspeed pitches, so models can become biased toward always predicting fastball.



Identity Confounding

Identity confounding: If the same pitchers are in both training and test sets, the model may memorize habits, boosting accuracy for the wrong reason.



Complexity & Reliability

The complexity of pitcher decisions makes full prediction reliability hard (impossible) to achieve. How then can we make reliable conclusions about pitcher's habits based on performance of our model?

Data Collection

 We wanted to allow one model to track identity

 Make list of pitchers

 80% of each pitcher's pitches in train set, 20% of each pitcher's pitches in test set

 Compare results with not tracking identity (tracking overarching distribution)

 Split list of pitchers into 80% of its volume and 20% of its volume

 80% list becomes train set, 20% list becomes test set

 No overlap - "subject-wise split"

Individual distributions vs. General distribution

Individual distributions – Each pitcher has a pitch-probability distribution.

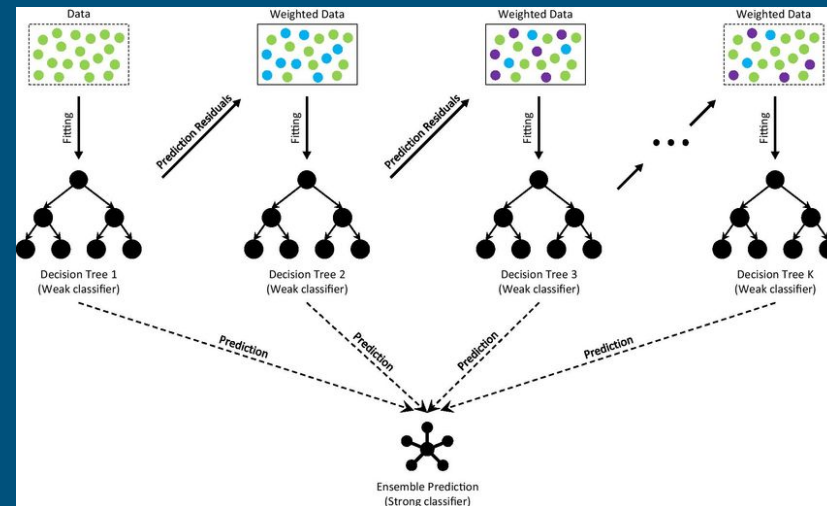
-  15 pitch types - each pitcher throws each with a probability
-  Identity tracking model forms these distributions in its memory

General distribution – An overarching general distribution of pitches exists.

-  15 pitch types - each pitch is thrown a certain amount of times in a year
-  Number of times pitch is thrown/total amount of pitches thrown
-  Non identity tracking model only has this distribution to go off of

Model Selection

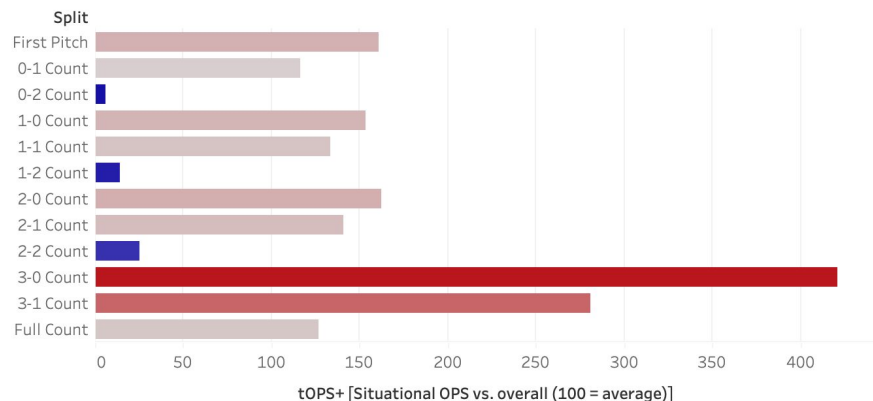
- Chosen Model: XGBoost Classifier
- Handles class imbalance
- Leverages gradient boosted trees to improve on weak learners
- Outperformed Logistic Regression, K-NN, Random Forest in baseline test



Data Preprocessing

- Score Differential:
 - Hitting team score - pitching team score
- Count Bucket:
 - Pitcher, hitter, neutral, full
 - Based on tOPS+
- Encoding:
 - Baserunners
 - Pitcher/Batter R or L

tOPS+ by Count



Model Construction



Pitcher-specific model:



Loop: one XGBoost model per pitcher



80/20 split within each pitcher's data



Global model



Single XGBoost model



80/20 random split across entire dataset



No pitcher in both train and test



Hyperparameter Tuning



Scale Positive Weight Multiplier



Number of Estimators



Learning Rate

Pitcher-Specific Model

```
#calculate the weight for fastball prediction
pos = y_tr.sum()
neg = len(y_tr) - pos
scale_pos_weight = (pos / neg) * 0.7

#xgboost model
clf = XGBClassifier(
    n_estimators=150,
    learning_rate=0.01,
    eval_metric='logloss',
    scale_pos_weight=scale_pos_weight,
    verbosity=0
)

#calculate weight for fastball prediction
pos = y_tr.sum()
neg = len(y_tr) - pos
scale_pos_weight = (pos / neg) * 0.7

#xgboost model
clf = XGBClassifier(
    n_estimators=200,
    learning_rate=0.01,
    eval_metric='logloss',
    scale_pos_weight=scale_pos_weight,
    verbosity=0
)
```

Non-Confounding Model

Accuracy Metrics and Results



Pitcher Specific:



Overall: 61.6%



Fastball: 62.6%



Offspeed: 60.3%



Global:



Overall: 57.7%



Fastball: 60.7%



Offspeed: 53.8%

Confusion Matrix - Pitcher Specific Model

cm	Predicted Fastball	Predicted Offspeed
Actual Fastball	46,834	28,023
Actual Offspeed	23,198	35,214

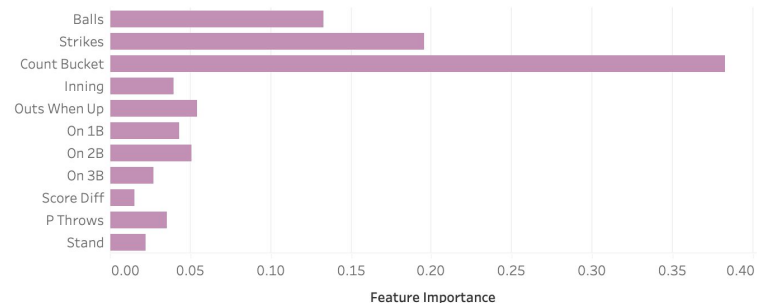
Confusion Matrix - Global Model

cm	Predicted Fastball	Predicted Offspeed
Actual Fastball	46,384	28,954
Actual Offspeed	26,984	29,704

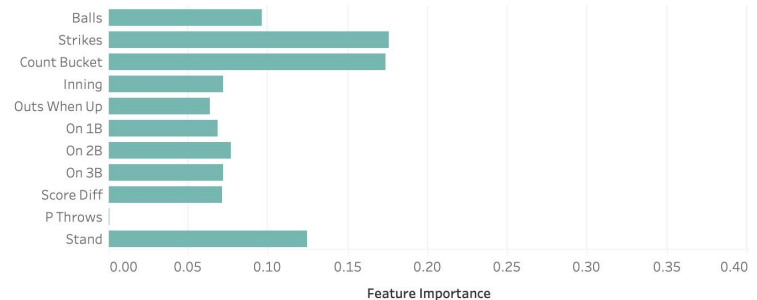
Feature Importance

- Ranked by total contributions across trees
- Top Features: balls, strikes, count bucket
- Global Model: top three 70% of importance weight
- Pitcher Specific Model: no feature > 18%
- Differences:
 - Count bucket
 - Reliance on league-wide trends

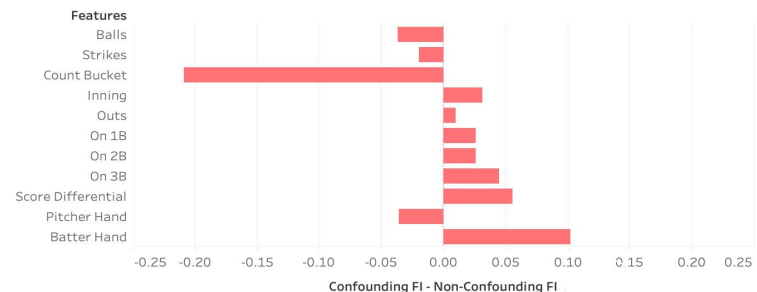
Identity Non-Confounding Model Feature Importance



Identity Confounding Model Feature Importance



Feature Importance Difference (Confounding-Non Confounding)



Another Interesting Result

Where the models went wrong:



Identity tracker: first pitches of an at-bat



Non-identity tracker: pitcher's counts



What is the underlying reason behind this?

- We suspected it was the difference between learning the general distribution and the pitcher's individual distributions

Likely cause of errors

Identity tracking model

 Likely errs most when pitchers stray from what they usually throw

Non-identity tracking model

 Likely errs most when pitchers stray from what a pitcher usually throws

Study of variation

Identity tracking model

-  Pitchers vary their pitches 29.9% differently (less variation) from their average distributions on first pitch situations
 -  On pitcher's counts, only 22.2% differently (less variation)

Non-identity tracking model

-  General distribution exhibits 10% more variation on pitcher's count situations with men on base than its average distribution
 -  On first pitches of at-bats, 8% less variation

Evidence so far:

-  Identity tracking: pitchers throw most differently from their usual selves in first pitch situations
 -  Exhibit less variation than they usually do in first pitch situations
-  Non-identity tracking: pitchers throw most differently from a typical pitcher in pitcher's count scenarios
 -  Exhibit more variation

What can we conclude?

Pitchers tap into their more obscure, personally preferred pitches in pitcher's count scenarios, but throw a more regimented set of “accepted” early-count pitches (tap into their more mainstream set of pitches)

 This would explain the differing variations of the distributions

 And would explain our models' errors

Moving forward:



Do more quantitative analysis of our final conclusion



Although difficult



Taking a pitcher's perspective

Thank you

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